Choose correct statement/s about anemia of chronic diseases

Select one or more:

- a. It is associated with decreased plasma concentrations of ferritin and transferrin
- b. Hepcidin level increases due to pro-inflammatory mediators ✓
- c. Patients usually have severe iron deficiency and very low plasma iron level
- d. IL 2 and IL 8 play a major role in the pathogenesis of this anemia
- e. It can be caused by Crohn's disease ✓

The correct answers are: It can be caused by Crohn's disease, Hepcidin level increases due to pro-inflammatory mediators

Choose correct statement/s about anemia of chronic diseases

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- a. It is associated with decreased plasma concentrations of ferritin and transferrin
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The correct answers are: It can be caused by Crohn's disease, Hepcidin level increases due to pro-inflammatory mediators

Question 8

Mark 1.00 out of 1.00

Choose the correct statement about pain

Select one or more:

- a. Disorders of pain perception include allodynia, paraparesis and analgesia
- b. Hypoxia of the internal organs can cause visceral pain ✓
- c. Colicky pain is typical for GERD
- d. Trigeminal neuralgia usually causes nociceptive pain
- e. Pain is a subjective symptom ✓

The correct answers are: Hypoxia of the internal organs can cause visceral pain, Pain is a subjective symptom

Question 5

Mark 1.00 out of 1.00

Describe 4 important symptoms occurring in patients with severe sideropenic anemia.

The RBC in sideropenic anemia are microcytic and hypochromic what lead to -

1. pallor - paleness of the skin deu to the low hemoglobin levels
2. fatigue - due to low oxygen supply to the tissues --> low metabolism
3. hypoxia - deu to low oxygen delivery to the tissues
4. weakness, tiredness - due to low oxygen supply to the tissues --> low metabolism

Comment:

Question 3

Mark 1.00 out of 1.00

Imagine you are in a simulation center and there is a "patient" in front of you with abdominal pain. The anamnesis corresponds to this. Write down major symptoms and important laboratory parameters, which you need to evaluate in the patient in order to exclude or support idea about inflammatory origin of pain.

laboratory parameters :

wcb count - amount of neutrophils, macrophages, mast cells, platlates.

ct or x -ray - to avaluate enlargment of an organ (spleen/liver/gall bladder/colon..)

patient recent history - constopation/diharria

stool sample - to evaluate infection / presence of blood

b12 / iron levels

symptoms:

nousia

vomitting

constipation

diharria

Question 2

Mark 0.00 out of 1.00

Iron deficiency

Select one or more:

- a. Can be caused by menorrhagias ✓
- b. Symptoms may include fatigue, paleness, dizziness ✓
- c. Usually leads to macrocytic anemia
- d. Usually leads to sideroblastic anemia ✗
- e. Can be caused by chronic low-grade blood loss from GIT

The correct answers are: Can be caused by chronic low-grade blood loss from GIT, Can be caused by menorrhagias, Symptoms may include fatigue, paleness, dizziness

Question 1

Mark 0.50 out of 1.00

Choose correct statement/s about pain

Select one or more:

- a. Colicky pain is usually mild and chronic
- b. Nociceptive pain arises from direct injury of axons of peripheral or central nerves
- c. Pain is objective
- d. Can be modulated by serotonin and noradrenalin systems ✓
- e. Can be caused by hypoxia

The correct answers are: Can be caused by hypoxia, Can be modulated by serotonin and noradrenalin systems

Question 2

Mark 0.50 out of 1.00

Gastroesophageal reflux disease is characterized by:

Select one or more:

- a. The highest incidence is in newborns leading to pylorospasm
- b. It is often treated by antibiotics
- c. Barret's esophagus is associated with it ✓
- d. It often leads to achalasia
- e. High prevalence in the adult population of developed countries

The correct answers are: High prevalence in the adult population of developed countries, Barret's esophagus is associated with it

Question 4

Mark 1.00 out of 1.00

Select correct statement\s about chronic gastritis type A

Select one or more:

- a. Patients usually have increased production of HCl and duodenal ulcers
- b. Patients often have anemia ✓
- c. Patients often have decreased level of HCl production ✓
- d. Patients have increased risk of development of stomach cancer ✓
- e. It usually develops in 2-3 days after acute gastritis

The correct answers are: Patients often have decreased level of HCl production, Patients often have anemia, Patients have increased risk of development of stomach cancer

Ketoacidosis occurs in patients with

Select one or more:

- a. simple starvation ✓
- b. acute liver failure
- c. alcohol abuse ✓
- d. uncontrolled type 1 diabetes ✓
- e. tissue hypoxia

The correct answers are: uncontrolled type 1 diabetes, simple starvation, alcohol abuse

Question 2

Mark 0.00 out of 1.00

The fecal occult blood test is a lab test used to check stool samples for hidden (occult) blood. The highest importance of this test is for diagnostics of the following disease out of the list:

Select one or more:

- a. Acute appendicitis
- b. Esophageal varices ✗
- c. Crohn's disease
- d. Hepatitis A
- e. Acute pancreatitis

The correct answer is: Crohn's disease

Question 4

Mark 1.00 out of 1.00

Describe pathogenesis of edema caused by increased hydrostatic pressure. Give 3 examples of causes.

increased hydrostatic pressure --> increased pressure on the walls of the capillary --> water moves from capillary to the interstitium --> water will not come back from the interstitium to venule end because of the high hydrostatic pressure

causes:

1. high blood pressure.
2. right heart failure
3.

Comment:

For classification of edema, following characterictic/s is/are used...

Select one or more:

- a. Pitting and non-pitting ✓
- b. Primary and secondary lymphedema
- c. Local and generalized ✓
- d. Acquired and inherited
- e. Hyperosmolar and hypoosmolar

The correct answers are: Local and generalized, Pitting and non-pitting, Primary and secondary lymphedema

In patients with hypertonic dehydration

Select one or more:

- a. it leads to an osmotic transfer of water from the cells into the interstitium
- b. decreased level of antidiuretic hormone can be the cause of the state ✓
- c. the volume of intracellular fluid increases ✗
- d. level of antidiuretic hormone is compensatory decreased
- e. the volume of intracellular fluids decreases

The correct answers are: the volume of intracellular fluids decreases, it leads to an osmotic transfer of water from the cells into the interstitium, decreased level of antidiuretic hormone can be the cause of the state

Question 6

Mark 1.00 out of 1.00

Explain the pathogenesis of Crohn's disease

crohns is one of the inflammatory bowel diseases. patient will develop ulcer.

in crohn's disease the most affected location is the ilium.

it affects all the layers of the mucosa in a granulomatous pattern.

it is immune related disease.

its not continous.

patient will have abdominal pain and nausea and diarreha.

Comment:

Question 8

Mark 1.00 out of 1.00

Write a definition of dehydration and give 6 examples of it (dividing into 3 subtypes).

dehydration = loss of body fluids.
can be hyperosmotic, hypoosmotic and isoosmotic.

1. excessive sweating.
2. diabetes incipidus.
3. no ADH secretion.
4. excessive urination.
5. diarriha.
6. vommiting.

Comment:

Select processes, which are common in the pathogenesis of gout:

Select one or more:

- [x] a. Hyperemia and severe pain of the affected joint during exacerbation ✓
- [x] b. High level of urea in blood ✗
- [x] c. With progression of gout, tophi appear on the hands, fingers, elbows, and ears ✓
- [x] d. Crystals of uric acid precipitate in the articular cavity ✓
- [ ] e. Myocardial infarction frequently is the initial sign of gout

The correct answers are: Crystals of uric acid precipitate in the articular cavity, Hyperemia and severe pain of the affected joint during exacerbation, With progression of gout, tophi appear on the hands, fingers,

Select correct statement/s about simple starvation:

Select one or more:

- [ ] a. Adult patients can usually survive more than 4 weeks, consuming only water
- [ ] b. During first 7 days of starvation, approximately 4-5 kilograms of body musculature is lost
- [ ] c. On the day 14, patient most probably will have metabolic alkalosis due to proteolysis
- [x] d. After several days, energy metabolism decreases ✓
- [ ] e. During first 10-12 hours, organism gets energy supply mostly due to lipolysis

The correct answers are: Adult patients can usually survive more than 4 weeks, consuming only water, After several days, energy metabolism decreases

Parents come to emergency care unit with 2-year-old boy after visiting their friends. They think that he drank there some unknown sweet fluid approximately 20 minutes ago, probably ethylene glycol. Ethylene glycol during its metabolism lead to formation of toxic oxalic acid. Which changes of acid base balance may develop, if the boy is not treated properly:

Select one or more:

- a. Metabolic alkalosis
- b. Respiratory acidosis
- c. High anion gap metabolic acidosis ✓
- d. Low or normal anion gap metabolic acidosis
- e. Most probably, no changes of acid base balance will develop

The correct answer is: High anion gap metabolic acidosis

The combination of hypokalemia and hyponatremia is characteristic for

Select one or more:

- a. acute consumption of large amounts of tap water in healthy person ✓
- b. In patient with diabetes insipidus
- c. primary aldosteronism
- d. adrenocortical insufficiency
- e. pure water dehydration

The correct answer is: acute consumption of large amounts of tap water in healthy person

Chose correct statement about H. pylory

Select one or more:

- a. Treatment of diseases, caused by H. pylory, includes antibiotics ✓
- b. Can be a cause of peptic ulcer of duodenum
- c. Can be a cause of inflammation of stomach ✓
- d. It is the leading cause of chronic hepatitis B
- e. Treatment of diseases, caused by H. pylory, includes corticosteroids

The correct answers are: Can be a cause of peptic ulcer of duodenum, Can be a cause of inflammation of stomach, Treatment of diseases, caused by H. pylory, includes antibiotics

Describe what is dysphagia and write down 6 important causes of dysphagia

Dysphagia = is difficulty swallowing

can be cased by:

1. Esophagial tumor
2. Nerve damage
3. Zenkers diverticulum
4. Stricture
5. Achalesia
6. Smoking
7. Low saliva prod.

Select correct answer/s:

Select one or more:

- [x] a. Patients with obesity are often characterized by decreased level of adiponectin ✔
- [ ] b. Adipocytes cannot secrete IL-1, TNF-alpha
- [ ] c. Insulin resistance affects glucose uptake by brain
- [x] d. Obesity is an important risk factor of hypertension ✔
- [x] e. Obesity can lead to restrictive disorder and respiratory insufficiency ✔

The correct answers are: Patients with obesity are often characterized by decreased level of adiponectin, Obesity is an important risk factor of hypertension, Obesity can lead to restrictive disorder and respiratory insufficiency

Obesity

Select one or more:

- [ ] a. Is defined as disease with excess of body fat and BMI above 25
- [x] b. Can lead to infertility ✔
- [x] c. Primary obesity is often associated by insulin resistance and leptin resistance ✔
- [x] d. Is a low grade inflammatory process, characterized by leukocytosis of 14*10^9/L or higher, increased sedimentation rate and several times increased CRP concentration in plasma ✖
- [x] e. Is a risk factor for accelerated development of atherosclerosis ✔

The correct answers are: Can lead to infertility, Is a risk factor for accelerated development of atherosclerosis, Primary obesity is often associated by insulin resistance and leptin resistance

Choose correct statement/s about hypercapnia

Select one or more:

- [ ] a. Hypercapnic acidosis causes vasodilation and increases systemic vascular resistance
- [ ] b. Hypercapnia may stimulate or depress the respiratory center
- [x] c. Severe hypercapnia may cause hypotension ✔
- [ ] d. Hypercapnic acidosis suppresses immune response
- [x] e. Chronic hypercapnia can be compensated by kidneys and usually well tolerated by patients ✔

The correct answers are: Hypercapnia may stimulate or depress the respiratory center, Hypercapnic acidosis suppresses immune response, Severe hypercapnia may cause hypotension, Chronic hypercapnia can be compensated by kidneys and usually well tolerated by patients

ARDS

Select one or more:

- [ ] a. Is often caused by left-side heart failure
- [x] b. Can lead to profound hypoxia and metabolic acidosis ✔
- [x] c. Can be a consequence of COVID-19 pneumonia ✔
- [x] d. Can be caused by inhalation of toxic chemicals ✔
- [ ] e. Is characterized by increased hydrostatic pressure and transudation to the alveoli

The correct answers are: Can be caused by inhalation of toxic chemicals, Can be a consequence of COVID-19 pneumonia, Can lead to profound hypoxia and metabolic acidosis

Question **6**

Select condition/s associated with hypercholesterolemia

Select one or more:

- [x] a. Anorexia nervosa ✔
- [ ] b. Type II diabetes mellitus
- [ ] c. Nephrotic syndrome
- [ ] d. Hypothyroidism
- [ ] e. Hepatocellular carcinoma with cachexia

The correct answers are: Type II diabetes mellitus, Anorexia nervosa, Nephrotic syndrome, Hypothyroidism

Question 7

Choose correct statement/s about iron:

Select one or more:

- [ ] a. Iron overload of the organism will never occur
- [x] b. Iron is absorbed mostly in the terminal (distal parts) of the ileum ✖
- [x] c. Iron deficiency often occurs in patients with chronic blood loss ✔
- [ ] d. In the case of multiple transfusions, iron may accumulate in the cells of the pancreas, liver, or heart
- [ ] e. Microcytic anemia, dysphagia, or angular stomatitis can be manifestations of iron deficiency

The correct answers are: In the case of multiple transfusions, iron may accumulate in the cells of the pancreas, liver, or heart, Iron deficiency often occurs in patients with chronic blood loss, Microcytic anemia, dysphagia, or angular stomatitis can be manifestations of iron deficiency

Choose correct statement/s about hypokalemia:

Select one or more:

- [ ] a. Muscle fatigue is one of the important features
- [x] b. It often develops in patients with severe metabolic acidosis ✖
- [ ] c. Plasma K+ concentrations are usually not lower than 15 mmol/L even in severe hypokalemia
- [ ] d. It is manifested on the ECG by high pointed T waves
- [ ] e. It can lead to a decrease in the reactivity of the renal tubules to ADH and to polyuria

The correct answers are: It can lead to a decrease in the reactivity of the renal tubules to ADH and to polyuria, Muscle fatigue is one of the important features

What applies to partially compensated metabolic acidosis

Select one or more:

- [ ] a. the pH value is always 7.40, because it is a compensated metabolic acidosis
- [ ] b. the pH is <7.0 because it is a metabolic acidosis
- [ ] c. BE (base excess) > 0 mmol/l
- [ ] d. the level of standard bicarbonates is high
- [x] e. BE (base excess) <-2.5 mmol/l ✔

The correct answer is: BE (base excess) <-2.5 mmol/l

Choose correct statement/s about chronic gastritis type A:

Select one or more:

- [ ] a. Patients usually have increased HCl production and duodenal ulcers
- [x] b. Patients have an increased risk of development of stomach cancer ✔
- [ ] c. It usually develops 2-3 weeks after acute gastritis
- [x] d. Patients often have reduced HCl production ✔
- [ ] e. Patients often have vitamin B12 deficiency

The correct answers are: Patients often have reduced HCl production, Patients often have vitamin B12 deficiency, Patients have an increased risk of development of stomach cancer

Severe diarrhea will most possibly lead to:

Select one or more:

- [x] a. Metabolic acidosis due to bicarbonate loss ✔
- [x] b. Dehydration ✔
- [ ] c. Metabolic acidosis due to compensatory HCl production
- [ ] d. Metabolic alkalosis due to compensatory hyperkalemia
- [ ] e. Metabolic alkalosis due to loss of bicarbonates

The correct answers are: Metabolic acidosis due to bicarbonate loss, Dehydration

Question 22

Choose correct statement/s about Crohn's disease

Select one or more:

- [ ] a. Extraintestinal complications include aphthae, eye inflammation
- [x] b. It can occur in any part of the digestive tract ✔
- [ ] c. Only women aged 20-30 suffer from Crohn's disease
- [ ] d. It is limited to the large intestine
- [ ] e. Common extraintestinal complications include hypovolemic shock

The correct answers are: It can occur in any part of the digestive tract, Extraintestinal complications include aphthae, eye inflammation

Ascites:

Select one or more:

- [x] a. It is accompanied by water retention ✔
- [ ] b. May be accompanied by hyponatremia due to hemodilution
- [ ] c. Is accompanied by decreased aldosterone secretion
- [ ] d. May develop due to a decrease in Na resorption in the proximal tubules
- [ ] e. May cause dyspnea in the patient

The correct answers are: May be accompanied by hyponatremia due to hemodilution, It is accompanied by water retention, May cause dyspnea in the patient

Prothrombin time (PT) increases in the case of

Select one or more:

- [x] a. Vitamin K deficiency ✔
- [ ] b. Thrombocytopenia
- [ ] c. Factor VII deficiency
- [ ] d. Marfan's syndrome
- [ ] e. Hemophilia A

The correct answers are: Factor VII deficiency, Vitamin K deficiency

Question **20**

Imagine the pathogenesis of sideropenic anemia and describe the manifestations of this disease, at the level of the whole organism (at least 1) and locally (3 manifestations are enough), eg on the skin/mucous membranes or the cardiovascular system.

intake disorders:

Select one or more:

- [x] A. Complications of mental anorexia include mild hypothyroidism and menstrual cycle disorders ✔
- [ ] B. Patients with mental bulimia always have decreased BMI (below 18.5 kg/m2)
- [ ] C. Mental anorexia is more common among men (age 15-25 y.o.) than woman (age 15-25 y.o.)
- [x] D. Orthorexia nervosa is an eating disorder characterized by an excessive preoccupation with eating healthy food ✔
- [x] E. Patients with mental bulimia often suffer from metabolic alkalosis ✔

The correct answers are: Complications of mental anorexia include mild hypothyroidism and menstrual cycle disorders, Patients with mental bulimia often suffer from metabolic alkalosis, Orthorexia nervosa is an eating disorder characterized by an excessive preoccupation with eating healthy food

Select correct statement/s about SIADH

Select one or more:

- [ ] A. In case of incorrect treatment pontine demyelinosis can occur
- [ ] B. It can cause hyponatremia
- [x] C. Swelling of lower extremities is common ✖
- [x] D. Possible cause of SIADH include ectopic production of ADH by lungs (for ex. lung carcinoma or pneumonia) ✔
- [ ] E. It is associated with decreased level of ANP

The correct answers are: It can cause hyponatremia, In case of incorrect treatment pontine demyelinosis can occur, Possible cause of SIADH include ectopic production of ADH by lungs (for ex. lung carcinoma or pneumonia)

During a visit do pediatrician, 14-year old girl is diagnosed with mental anorexia. Select signs and symptoms, which helped doctor to put this diagnosis:

Select one or more:

- [x] a. Body weigh below 5 per centile ✔
- [ ] b. Ascites due to hypoproteinemia
- [x] c. Absence of menses for more than 3 months after she had regular menses from 12 to 13 years ✔
- [ ] d. Body heights above 95 per centile
- [x] e. BMI 21,5 kg/m2 ✖

The correct answers are: Body weigh below 5 per centile, Absence of menses for more than 3 months after she had regular menses from 12 to 13 years

Unconscious young male is in an intensive care unit. His skin is pale, and extremities are cold, BMI is 23 kg/m2, arterial blood pressure 90/60 mm Hg, heart rate 130 bpm, body temperature 37.2 C, respiratory rate 28/min. Laboratory findings: red blood cells 5,8*10^12, Leukocytes 12*10^9, ESR 8 mm/h, Na+ 145 mmol/L, Cl 105 mmol/L, glucose level 22,2 mmol/L, HCO3 10 mmol/L, blood pH 7.10. Which conclusion/s can be made on the abovementioned information?

Select one or more:

- [ ] a. Most probably patient has diabetic ketoacidosis ✔
- [ ] b. Patient has hyperosmolar dehydration due to vomiting
- [ ] c. Patient has hyperosmolar dehydration due to sweating
- [ ] d. Patient has hyperosmolar dehydration probably to decompensation of diabetes mellitus ✔
- [ ] e. Patient has hyperosmolar dehydration due to diabetes insipidus

The correct answers are: Patient has hyperosmolar dehydration probably to decompensation of diabetes mellitus, Most probably patient has diabetic ketoacidosis

Write a definition of hyperhydration and give 6 examples of it (dividing into 3 subtypes).

hyper hydration is the increase hydration of a person due to several reasons

hypotonic hyper hydration - caused by drinking pure water, can be caused by hypotonic infusion therapy ,

isotonic hyper hydration - can be caused by isotonic infusion therapy , SIADH

hypertonic hyper hydration - can be caused by hypertonic infusion, increase amount of salty drinks,

Comment:

Bicarbonate deficiency can be caused by

Select one or more:

- [ ] a. severe diarrhea ✔
- [ ] b. hyperaldosteronism
- [ ] c. dilution when the amount of HCO3 is relatively decreased (e.g. after infusion of isotonic solutions without pufration bases) ✔
- [ ] d. renal tubular acidosis ✔
- [ ] e. none of the mentioned

The correct answers are: renal tubular acidosis, dilution when the amount of HCO3 is relatively decreased (e.g. after infusion of isotonic solutions without pufration bases), severe diarrhea

Ascites may be caused by

Select one or more:

- [ ] a. insufficient production of aldosterone
- [ ] b. severe loss of plasma proteins in patients with diseases of kidneys (nephrotic syndrome)
- [x] c. peritoneal carcinomatosis ✔
- [ ] d. increase of oncotic pressure
- [x] e. portal hypertension ✔

The correct answers are: portal hypertension, peritoneal carcinomatosis, severe loss of plasma proteins in patients with diseases of kidneys (nephrotic syndrome)

Select correct statement/s about stress starvation

Select one or more:

- [ ] a. Level of insulin is usually decreased
- [x] b. Characterized by intensive catabolism of proteins ✔
- [ ] c. Severe emotional stress (for ex. death of partner) is one of the most important causes
- [x] d. Level of glucose is often increased ✔
- [ ] e. Ketone bodies are often increased

The correct answers are: Characterized by intensive catabolism of proteins, Level of glucose is often increased

Mother of 8-month old boy comes to doctor complaining that her son does not seem to be well after she stopped breastfeeding and started to give him some cereals approximately 2 weeks ago. He had diarrhea several days intermittently, decreased appetite, stomachache and bloating, and lost approximately 0.5 kg of weight during 2 weeks. He did not vomit and did not have fever during last month. Before that episode seemed to be a generally healthy child. Most probably patient has:

Select one or more:

- [ ] a. Lactose intolerance
- [ ] b. Fructose intolerance
- [ ] c. Cystic fibrosis
- [x] d. Celiac disease ✔
- [ ] e. Phenilketonuria

The correct answer is: Celiac disease

During panic attack, young female complains on feeling on palpitation, lightheadedness, rapid and deep breathing, tingling in the arms. To decrease these symptoms, her friend recommends her to breathe into a paper bag during the attack. Select correct statement/s about the situation:

Select one or more:

- [x] a. Most probably, patient develops respiratory alkalosis during the panic attack ✔
- [ ] b. Lightheadedness in the patient is most probably related to vasodilation of the brain vessels
- [ ] c. Most probably, at least some of her symptoms are connected to decreased level of CO2
- [ ] d. Breathing to the paper bag is not helpful and she most probably needs to be hospitalized to and intensive care unit to prevent severe brain damage
- [ ] e. Most probably, patient develops metabolic alkalosis during the panic attack

The correct answers are: Most probably, patient develops respiratory alkalosis during the panic attack, Most probably, at least some of her symptoms are connected to decreased level of CO2

Select typical signs and symptoms of vitamin deficiencies:

Select one or more:

- [ ] a. Vitamin C deficiency severely increases risk of thrombosis
- [ ] b. Vitamin A deficiency leads to night blindness, xerophtalmia, keratinization of the skin and of the mucous membranes ✔
- [ ] c. Vitamin D deficiency increases risk of muscle weakness, muscle cramps ✔
- [ ] d. Taking broad-spectrum antibiotics long term can lead to deficiency of vitamin K and increase risk of bleeding
- [ ] e. Vitamin D deficiency increases risk of mood changes and depression

The correct answers are: Vitamin A deficiency leads to night blindness, xerophtalmia, keratinization of the skin and of the mucous membranes, Vitamin D deficiency increases risk of muscle weakness, muscle cramps, Vitamin D deficiency increases risk of mood changes and depression, Taking broad-spectrum antibiotics long term can lead to deficiency of vitamin K and increase risk of bleeding